

① Chapter 10. RATE DISTORTION THEORY

- The description of an arbitrary real number.
- "goodness" of a representation of a source
- distortion measure which is a measure of distance between the random variable and its representation.

② 10.1 QUANTIZATION

How complicated it is to solve the quantization problem exactly for a single random variable.

Let the random variable be represented by X . $\rightarrow R$ bits

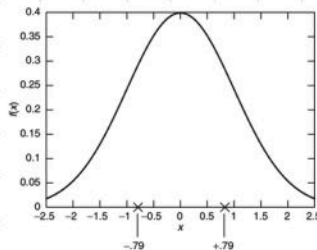
Let the representation of X be denoted as $\hat{X}(X)$ $\rightarrow 2^R$ values

$\Rightarrow$ Problem: Find the optimum set of values of $\hat{X}$ (called **reproduction points** or **code points**) and regions that are associated with each value $\hat{X}$.

Example) $X \sim N(0, \sigma^2)$ Assume a squared-error distribution measure $\min \mathbb{E}(X - \hat{X}(X))^2$

If 1 bit $\Rightarrow$ distinguish whether or not $X > 0$

$$\hat{X}(x) = \begin{cases} \sqrt{\frac{2}{\pi}} \sigma & \text{if } x \geq 0 \\ -\sqrt{\frac{2}{\pi}} \sigma & \text{if } x < 0 \end{cases}$$



③

If 2 bits $\Rightarrow$ not as simple

(divide the real line into four regions)

Two simple property of optimal regions and reconstruction point

- Given a set $\{\hat{X}(w)\}$ of reconstruction points, the distortion is minimized by mapping a source random variable X to the representation $\hat{X}(w)$ that is closed to it. The set of regions of X defined by this mapping is called a **Voronoi** and **Dirichlet partition** defined by the reconstruction points
- The reconstruction points should minimize the conditional expected distortion over their respective assignment regions

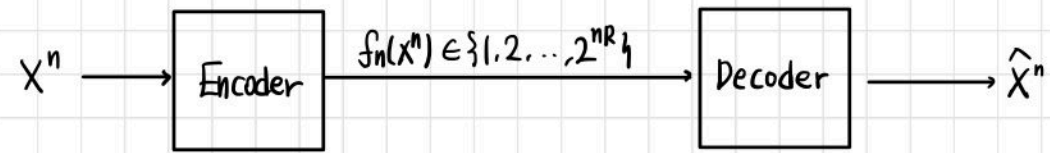
④ These two properties enable us to construct a simple algorithm to find a "good" quantizer.

Lloyd Algorithm (frequently used to design quantization system)

- ① start with a set of reconstruction points
- ① find the optimal set of reconstruction regions
- ② find the optimal reconstruction points for these regions
- ④ repeat the iteration for this new set of reconstruction points

⑦ 10.2. DEFINITIONS

Assume $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} p(x), x \in \mathcal{X}$



Def) distortion function or distortion measure d

$$d: \mathcal{X} \times \hat{\mathcal{X}} \rightarrow \mathbb{R}^+$$

Def) d is bounded if $d_{\max} := \max_{x \in \mathcal{X}, \hat{x} \in \hat{\mathcal{X}}} d(x, \hat{x}) < \infty$

⑥ Examples of distortion functions

1. Hamming distortion

$$d(x, \hat{x}) = \begin{cases} 0 & \text{if } x = \hat{x} \\ 1 & \text{if } x \neq \hat{x} \end{cases} \quad \Rightarrow E d(X, \hat{X}) = P(X \neq \hat{X})$$

2. Square-error distribution

$$d(x, \hat{x}) = (x - \hat{x})^2$$

Advantages: simplicity, relationship to least-squares prediction

But in applications (as image, speech coding)

it is not an approximate measure of distortion for human observers.

①

Def) distortion between sequences x^n and $\hat{x}^n$

$$d(x^n, \hat{x}^n) = \frac{1}{n} \sum_{i=1}^n d(x_i, \hat{x}_i)$$

* This is not the only reasonable definition.

(For example, we can use $\max \{ d(x_i, \hat{x}_i) \}$)

Def) $(2^{nR}, n)$ -rate distortion code

- encoding function $f_n: x^n \rightarrow \{1, 2, \dots, 2^{nR}\}$
- decoding function $g_n: \{1, 2, \dots, 2^{nR}\} \rightarrow \hat{x}^n$
- distortion associated with the $(2^{nR}, n)$: $D = \mathbb{E} d(X^n, g_n(f_n(X^n)))$
- **codebook**: set of n -tuples $g_n(1), \dots, g_n(2^{nR})$ denoted by $\hat{x}^n(1), \dots, \hat{x}^n(2^{nR})$
- **assignment regions**: $f_n^{-1}(1), \dots, f_n^{-1}(2^{nR})$
- $\hat{x}^n$: vector quantization, reproduction, reconstruction, representation, source code, estimate

⑧

Def) rate distortion pair (R, D) : **achievable**

if $\exists$ sequence of $(2^{nR}, n)$ -rate distortion codes (f_n, g_n) with $\lim_{n \rightarrow \infty} \mathbb{E} d(X^n, g_n(f_n(X^n))) \leq D$

Def) **rate distortion region** for a source is the closure of the set of achievable rate distortion pairs (R, D)

Def) **rate distortion function** $R(D)$ is the infimum of rates R s.t. (R, D) is in the rate distortion region of the source for a given distortion D

Def) **distortion rate function** $D(R)$ is the infimum of all distortions D s.t. (R, D) is in the rate distortion region of the source for a given rate R .

⑨ Def) information rate distortion function $R^{(I)}(D)$ for a source X with distribution measure $d(x, \hat{x})$

$$R^{(I)}(D) = \min_{p(\hat{x}|x) : \sum_{(x, \hat{x})} p(x) p(\hat{x}|x) d(x, \hat{x}) \leq D} I(X; \hat{X})$$

Thm 10.2.1) rate distortion function for iid source $X (\sim p(x))$ and bounded $d(x, \hat{x})$ is equal to associated information rate distortion function. Thus,

$$R(D) = R^{(I)}(D)$$

is the minimum achievable rate at distortion D .

Hence we will use $R(D)$ from now.

Before coming to the proof, calculating...!

⑩ 10.3 CALCULATION OF THE RATE DISTORTION FUNCTION

10.3.1 Binary Source.

Thm 10.3.1) rate distortion function for a Bernoulli(p) source with Hamming distortion is given by

$$R(D) = \begin{cases} H(p) - H(D) & 0 \leq D \leq \min\{p, 1-p\} \\ 0 & D > \min\{p, 1-p\} \end{cases}$$

sketch of proof)

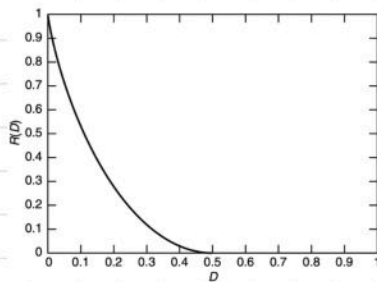
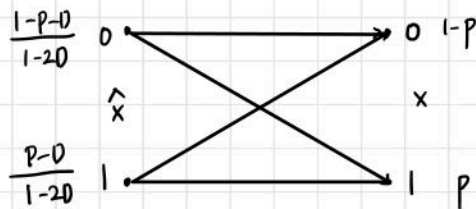
$$R(D) = \min I(X, \hat{X})$$

$$\begin{aligned} I(X, \hat{X}) &= H(X) - H(X|\hat{X}) \\ &= H(p) - H(X \oplus \hat{X} | \hat{X}) \end{aligned}$$

$$\geq H(p) - H(X \oplus \hat{X})$$

$$\geq H(p) - H(D)$$

$$\Rightarrow R(D) \geq H(p) - H(D)$$



⑪ 10.3.2 Gaussian Source

Thm 10.3.2) rate distortion function for a $N(0, \sigma^2)$ source with squared-error distortion is

$$R(D) = \begin{cases} \frac{1}{2} \log \frac{\sigma^2}{D} & , 0 \leq D \leq \sigma^2 \\ 0 & D > \sigma^2 \end{cases}$$

sketch of pf)

$$R(D) = \min_{f(\hat{x}|x) : E(\hat{x}-x)^2 \leq D} I(X; \hat{X})$$

$$\Leftrightarrow D(R) = \sigma^2 2^{-2R}$$

$$I(X; \hat{X}) = h(X) - h(X|\hat{X})$$

$$= \frac{1}{2} \log(2\pi e) \sigma^2 - h(X - \hat{X}|\hat{X})$$

$$\geq \frac{1}{2} \log(2\pi e) \sigma^2 - h(X - \hat{X})$$

$$\geq \frac{1}{2} \log(2\pi e) \sigma^2 - h(N(0, E(X - \hat{X})^2))$$

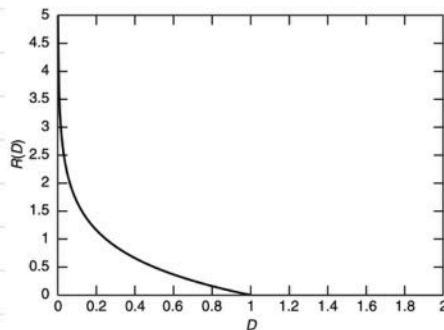
$$= \frac{1}{2} \log(2\pi e) \sigma^2 - \frac{1}{2} \log(2\pi e) E(X - \hat{X})^2$$

$$\geq \frac{1}{2} \log(2\pi e) \sigma^2 - \frac{1}{2} \log(2\pi e) D$$

$$= \frac{1}{2} \log \frac{\sigma^2}{D}$$

$$\hat{X} \sim N(0, \sigma^2 - D) \rightarrow \oplus \rightarrow X \sim N(0, \sigma^2)$$

↑
 $Z \sim N(0, D)$



⑫ 10.3.3 Simultaneous Description of Independent Gaussian Random Variables

Thm 10.3.3) (Rate distortion for a parallel Gaussian source)

Let $X_i \sim N(0, \sigma_i^2)$ $i=1, 2, \dots, m$ be independent Gaussian random variables,

and let the distortion measure be $d(x^m, \hat{x}^m) = \sum_{i=1}^m (x_i - \hat{x}_i)^2$

Then the rate distortion function is given by

$$R(D) = \sum_{i=1}^m \frac{1}{2} \log \frac{\sigma_i^2}{D_i}$$

where
$$D_i = \begin{cases} \lambda & \text{if } \lambda < \sigma_i^2 \\ \sigma_i^2 & \text{if } \lambda \geq \sigma_i^2 \end{cases}$$

where λ is chosen so that $\sum_{i=1}^m D_i = D$

sketch of pf)
$$I(x^m, \hat{x}^m) \geq \sum_{i=1}^m \left(\frac{1}{2} \log \frac{\sigma_i^2}{D_i} \right)^+$$

Using Lagrange multiplier $\Rightarrow \square$

(13)

10.4. CONVERSE TO THE RATE DISTORTION THEOREM

Lemma 10.4.1) (Convexity of $R(D)$) $R(D)$ is a nonincreasing convex function.

$$R(D) = \min_{p(\hat{x}|x): \sum_{(\hat{x}, x)} p(x)p(\hat{x}|x)d(x, \hat{x}) \leq D} I(X; \hat{X})$$

pf) $\int p(x)p(\hat{x}|x)d(x, \hat{x})$ is increasing as D increases

$\Rightarrow R(D)$ is nonincreasing in D .

Consider (R_1, D_1) , (R_2, D_2) , joint distributions: $p_1(x, \hat{x}) = p(x)p_1(\hat{x}|x)$, $p_2(x, \hat{x}) = p(x)p_2(\hat{x}|x)$

Consider $p_\lambda = \lambda p_1 + (1-\lambda)p_2$. Since distortion function is linear of distribution

$\Rightarrow D(p_\lambda) = \lambda D_1 + (1-\lambda)D_2$, Mutual Information is convex of conditional dist (Thm 2.7.4)

$$\text{Hence } I_{p_\lambda}(X; \hat{X}) \leq \lambda I_{p_1}(X; \hat{X}) + (1-\lambda)I_{p_2}(X; \hat{X})$$

$$\therefore \text{By def } R(D_\lambda) \leq I_{p_\lambda}(X; \hat{X})$$

$$\leq \lambda I_{p_1}(X; \hat{X}) + (1-\lambda)I_{p_2}(X; \hat{X})$$

$$= \lambda R(D_1) + (1-\lambda)R(D_2)$$

(14) Converse of Thm 10.2.1 proof)

We must show for any source X drawn (iid $p(x)$) with $d(x, \hat{x})$, any $(2^{nR}, n)$ rate distortion code with distortion $\leq D$, that the rate R of the code satisfies $R \geq R(D)$.

Let $f_n: X^n \rightarrow \{1, 2, \dots, 2^{nR}\}$, $g_n: \{1, 2, \dots, 2^{nR}\} \rightarrow \hat{X}^n$

Let $\hat{X}^n = \hat{X}^n(X^n) = g_n(f_n(X^n))$ be the reproduced sequence corresponding to X^n .

then we have following chain of inequality.

$$nR \geq H(f_n(X^n)) \geq H(f_n(X^n)) - H(f_n(X^n) | X^n) = I(X^n; f_n(X^n)) \geq I(X^n; \hat{X}^n)$$

$$= H(X^n) - H(X^n | \hat{X}^n) = \sum_{i=1}^n H(X_i) - H(X^n | \hat{X}^n) = \sum_{i=1}^n H(X_i) - \sum_{i=1}^n H(X_i | \hat{X}^n, X_{i-1}, \dots, X_1)$$

$$\geq \sum_{i=1}^n H(X_i) - \sum_{i=1}^n H(X_i | \hat{X}_i) = \sum_{i=1}^n I(X_i; \hat{X}_i) \geq \sum_{i=1}^n R(E_d(X_i, \hat{X}_i)) = n \left(\frac{1}{n} \sum_{i=1}^n R(E_d(X_i, \hat{X}_i)) \right)$$

$$\geq nR \left(\frac{1}{n} \sum_{i=1}^n E_d(X_i, \hat{X}_i) \right) = nR(E_d(X^n, \hat{X}^n)) = nR(D)$$

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⑫ 10.5 ACHIEVABILITY OF THE RATE DISTORTION FUNCTION

Def) $p(x, \hat{x})$: joint probability distribution on $\mathcal{X} \times \hat{\mathcal{X}}$, $d(x, \hat{x})$: distortion measure

For any $\varepsilon > 0$, a pair of sequences $(x^n, \hat{x}^n)$ is **distortion ε -typical**, **distortion typical** if

$$|-\frac{1}{n} \log p(x^n) - H(X)| < \varepsilon, \quad |-\frac{1}{n} \log p(\hat{x}^n) - H(\hat{X})| < \varepsilon$$

$$|-\frac{1}{n} \log p(x^n, \hat{x}^n) - H(X, \hat{X})| < \varepsilon, \quad |d(x^n, \hat{x}^n) - \mathbb{E}d(X, \hat{X})| < \varepsilon$$

The set of distortion typical sequences is called the **distortion typical set**, $A_{d, \varepsilon}^{(n)}$

Lemma 10.5.1) $(X_i, \hat{X}_i)$: drawn iid $p(x, \hat{x}) \Rightarrow \Pr(A_{d, \varepsilon}^{(n)}) \rightarrow 1$ as $n \rightarrow \infty$

pf) By Law of large number

(16) Lemma 10.5.2) For all $(x^n, \hat{x}^n) \in A_{\delta, \varepsilon}^{(n)}$

$$P(\hat{x}^n) \geq P(\hat{x}^n | x^n) 2^{-n(I(x; \hat{x}) + 3\varepsilon)}$$

$$\begin{aligned} \text{pf)} \quad P(\hat{x}^n | x^n) &= \frac{P(x^n, \hat{x}^n)}{P(x^n)} = P(\hat{x}^n) \cdot \frac{P(x^n, \hat{x}^n)}{P(x^n)P(\hat{x}^n)} \\ &\leq P(\hat{x}^n) \frac{2^{-n(H(x, \hat{x}) - \varepsilon)}}{2^{-n(H(x) + \varepsilon)} 2^{-n(H(\hat{x}) + \varepsilon)}} = P(\hat{x}^n) 2^{n(I(x; \hat{x}) + 3\varepsilon)} \end{aligned}$$

Lemma 10.5.3) For $0 \leq x, y \leq 1$, $n > 0$

$$(1 - xy)^n \leq 1 - x + e^{-y^n}$$

$$\text{pf)} \quad \text{Using } f(y) = e^{-y} - 1 + y \quad \Rightarrow \quad (1 - y)^n \leq e^{-y^n}$$

$$\begin{aligned} \text{Using } g_y(x) = (1 - xy)^n \text{ \& Jensen} \quad \Rightarrow \quad (1 - xy)^n &= g_y(x) \leq (1 - x)g_y(0) + xg_y(1) \\ &\leq 1 - x + e^{-y^n} \end{aligned}$$

(17) Achievability in Thm 10.2.1)

$X_1 \cdots X_n \sim \text{iid } p(x)$, $d(x, \hat{x})$: bound distortion measure, $R(D)$: rate distortion ft.

For any D , any $R > R(D)$, WANT (R, D) : achievable

By proving existence of a seq of rate distortion codes with rate R , distortion D .

Fix $p(\hat{x}|x)$ ($p(\hat{x}|x)$ achieves $\min_{p(\hat{x}|x): \sum_{x, \hat{x}} p(x)p(\hat{x}|x)d(x, \hat{x}) \leq D} I(x; \hat{x}) \Rightarrow I(x; \hat{x}) = R(D)$)

$$p(\hat{x}) = \sum_x p(x)p(\hat{x}|x)$$

(choose $\delta > 0$)

WANT existence of a rate distortion code with rate R , distortion $\geq D + \delta$

(18)

Generation of codebook : Randomly generate a rate distortion codebook $\mathcal{C}$ consisting of 2^{nR} sequences $\hat{x}^n$ drawn i.i.d $\prod_{i=1}^n p(\hat{x}_i)$, Index : $\omega = \{1, 2, \dots, 2^{nR}\}$

Encoding : Encode x^n by ω if there exists a ω s.t. $(x^n, \hat{x}^n(\omega)) \in A_{d, \epsilon}^{(n)}$

(If there is more than one $\Rightarrow$ send the least)

(If there is no such ω , let $\omega=1$)

$\Rightarrow nR$ bits suffice to describe the index ω of the jointly typical codeword.

Decoding : reproduced seq in $\hat{x}^n(\omega)$

Calculating of distortion

$$\bar{D} = E_{x^n, c} d(x^n, \hat{x}^n)$$

(E is over the random choice of codebooks and over x^n)

(a) For fixed codebook $\mathcal{C}$, choice of $\epsilon > 0$, divide $x^n \in \mathcal{X}^n$ into two category

① x^n s.t. $\exists \hat{x}^n(\omega)$ that $d(x^n, \hat{x}^n(\omega)) < D + \epsilon$.

Since the total probability of these seq is at most 1, these seq contribute at most $D + \epsilon$ to the expected distortion.

② x^n s.t. $\nexists \hat{x}^n(\omega)$ that $d(x^n, \hat{x}^n(\omega)) < D + \epsilon$

P_e := total probability of these seq

Since distortion for any seq $\leq d_{\max}$, these seq contribute at most $P_e d_{\max}$ to the expected distortion

$$\Rightarrow \mathbb{E} d(x^n, \hat{x}^n(\omega)) \leq D + \epsilon + P_e d_{\max}$$

If P_e is small enough, can be made less than $D + \delta$ for choice of ϵ

WTS) P_e is small

(20) calculating of P_e

Let $J(C)$: set of source seq x^n s.t. at least one codeword in C is distortion typical with x^n .

$$\Rightarrow P_e = \sum_C P(C) \sum_{x^n: x^n \notin J(C)} P(x^n)$$

$$\text{Let } K(x^n, \hat{x}^n) = \begin{cases} 1 & \text{if } (x^n, \hat{x}^n) \in A_{d, \epsilon}^{(n)} \\ 0 & \text{if } (x^n, \hat{x}^n) \notin A_{d, \epsilon}^{(n)} \end{cases}$$

$$\Rightarrow = \sum_{x^n} P(x^n) \sum_{C: x^n \in J(C)} P(C)$$

$$\text{For fixed } x^n; \quad P((x^n, \hat{x}^n) \notin A_{d, \epsilon}^{(n)}) = P(K(x^n, \hat{x}^n) = 0) = 1 - \sum_{\hat{x}^n} P(\hat{x}^n) K(x^n, \hat{x}^n)$$

$$\Rightarrow = \sum_{x^n} P(x^n) \left[1 - \sum_{\hat{x}^n} P(\hat{x}^n) K(x^n, \hat{x}^n) \right]^{2^{nR}}$$

$$(\text{by Lemma 10.5.2} \quad \sum_{\hat{x}^n} P(\hat{x}^n) K(x^n, \hat{x}^n) \geq \sum_{\hat{x}^n} P(\hat{x}^n | x^n) 2^{-n(I(x; \hat{x}) + 3\epsilon)} K(x^n, \hat{x}^n))$$

$$\Rightarrow P_e \leq \sum_{x^n} P(x^n) \left(1 - 2^{-n(I(x; \hat{x}) + 3\epsilon)} \sum_{\hat{x}^n} P(\hat{x}^n | x^n) K(x^n, \hat{x}^n) \right)^{2^{nR}}$$

(2)

$$\begin{aligned}
 P_e &\leq \frac{\sum_{x^n} P(x^n) \left(1 - 2^{-n(I(X;\hat{X}) + 3\varepsilon)} \sum_{\hat{x}^n} P(\hat{x}^n | x^n) K(x^n, \hat{x}^n) \right)^{2^{nR}}}{\underbrace{\sum_{\hat{x}^n} P(\hat{x}^n | x^n) K(x^n, \hat{x}^n)}_{=P} + \underbrace{e^{-\left(2^{-n(I(X;\hat{X}) + 3\varepsilon)}\right) 2^{nR}}}_{\rightarrow 0 \text{ as } n \rightarrow \infty \text{ (if } R > I(X;\hat{X}) + 3\varepsilon)}} \\
 &= P
 \end{aligned}$$

(We have $R > R(0)$) $\Rightarrow R > I(X;\hat{X})$ $\Rightarrow \text{Can choose } \varepsilon$

$$\begin{aligned}
 &\leq \frac{1 - \sum_{x^n} \sum_{\hat{x}^n} P(x^n) P(\hat{x}^n | x^n) K(x^n, \hat{x}^n)}{1 - \sum_{x^n} \sum_{\hat{x}^n} P(x^n, \hat{x}^n) K(x^n, \hat{x}^n)} \\
 &= \Pr((X^n, \hat{X}^n) \notin A_{d,\varepsilon}^{(n)}) < \varepsilon \quad \text{for } n \text{ suff large}
 \end{aligned}$$

(22) Existence of a rate distortion code with an expected distortion close to D and a rate closed to $R(D)$.

random coding proof of the channel coding thm

$\Leftrightarrow$ random coding proof of the rate distortion thm.
evident

② Channel coding for the Gaussian channel

$Y_i = X_i + Z_i$, $Z_i \sim \text{iid } N(0, N)$, P : power constraint on the power per symbol of the transmitted codeword.

Consider seq of n transmissions, transmitted seq $\in$ sphere of radius $\sqrt{nP}$ in $\mathbb{R}^n$

Coding problem $\Leftrightarrow$ find a set of 2^{nR} seqs within this sphere

s.t. prob of mistake is small (the sphere of radius $\sqrt{nN}$ is almost disjoint)

$\Rightarrow$ filling a $\sqrt{n(P+N)}$ sphere with $\sqrt{nN}$ spheres

M : the number of codewords that can be transmitted efficiently

$$\Rightarrow M \leq \frac{(\sqrt{n(P+N)})^n}{(\sqrt{nN})^n} = \left(\frac{P+N}{N}\right)^{\frac{n}{2}} \cong 2^{nC}$$

$\uparrow$ Channel coding thm.

②4 ✓ Rate distortion for the Gaussian source

Consider Gaussian source of variance σ^2

$(2^{nR}, n)$ rate distortion code for this source, distortion D

$\mathcal{D}$ is a set of 2^{nR} sequences in $\mathbb{R}^n$ s.t. most source seq of length n are
within a distance $\sqrt{nD}$ of some codeword. ($\in \sqrt{n\sigma^2}$ sphere)

By sphere packing, $2^{nR(D)} = \left(\frac{\sigma^2}{D}\right)^{n/2}$

The rate distortion thm shows the minimum rate is asymptotically achievable.

(i.e. that there exists collection of sphere of radius $\sqrt{nD}$ that cover the space except for a set of arbitrary small probability)

②5) 10.6 STRONGLY TYPICAL SEQUENCES AND RATE DISTORTION.

Def) A sequence $x^n \in \mathcal{X}^n$ is said to be ε -strongly typical w.r.t a distribution $p(x)$ on $\mathcal{X}$ if

1. For all $a \in \mathcal{X}$ with $p(a) > 0$, we have

$$\left| \frac{1}{n} N(a|x^n) - p(a) \right| < \frac{\varepsilon}{|\mathcal{X}|}$$

2. For all $a \in \mathcal{X}$ with $p(a) = 0$, $N(a|x^n) = 0$

($N(a|x^n)$ is number of occurrences of the symbol a in the sequence x^n .)

$A_{\varepsilon}^{*(n)}(\mathcal{X})$ or $A_{\varepsilon}^{*(n)}$: strongly typical set : set of sequences $x^n \in \mathcal{X}^n$ s.t. x^n is strongly typical

(26)

Def) A pair of sequences $(x^n, y^n) \in X^n \times Y^n$ is ε -strongly typical w.r.t $p(x, y)$ on $X \times Y$ if

1. For all $(a, b) \in X \times Y$ with $p(a, b) > 0$, we have

$$\left| \frac{1}{n} N(a, b | x^n, y^n) - p(a, b) \right| < \frac{\varepsilon}{|X||Y|}$$

2. For all $(a, b) \in X \times Y$ with $p(a, b) = 0$,

$$N(a, b | x^n, y^n) = 0$$

($N(a, b | x^n, y^n)$ is the number of occurrences of the pair (a, b) in the pair of sequences (x^n, y^n) .)

$A_{\varepsilon}^{*(n)}(X, Y)$, $A_{\varepsilon}^{*(n)}$: strongly typical set: set of sequences $(x^n, y^n) \in X^n \times Y^n$ s.t. (x^n, y^n) is strongly typical

Note) By def $(x^n, y^n) \in A_{\varepsilon}^{*(n)}(X, Y) \Rightarrow x^n \in A_{\varepsilon}^{*(n)}(X)$

② Lemma 10.6.1) Let (X_i, Y_i) be drawn i.i.d $\sim p(x, y)$.

Then $\Pr(A_{\varepsilon}^{*(n)}) \rightarrow 1$ as $n \rightarrow \infty$

proof) From the strong law of Large numbers, the Lemma is immediate.

Lemma 10.6.2) [Extension of thm 7.6.1]

Let $Y_1, \dots, Y_n \sim \text{i.i.d } p(y)$, $x^n \in A_{\varepsilon}^{*(n)}(X)$

$$\Rightarrow 2^{-n(I(x;Y) + \varepsilon_1)} \leq \Pr((x^n, Y^n) \in A_{\varepsilon}^{*(n)}) \leq 2^{-n(I(x;Y) - \varepsilon_1)}$$

where ε_1 goes to 0 as $\varepsilon \rightarrow 0$ and $n \rightarrow \infty$

proof) We will not prove

(28) Proceed directly to the achievability of the rate distortion function

It is similar to 10.5 proof

But we divide the seq $x^n \in \mathcal{X}^n$ into three categories

- Nontypical seq $x^n \notin A_{\epsilon}^{*(n)}$

- Typical seq $x^n \in A_{\epsilon}^{*(n)}$ s.t. $\exists$ codeword $\hat{x}^n(w)$ that is jointly typical with x^n .

- Typical seq $x^n \in A_{\epsilon}^{*(n)}$ s.t. $\nexists$ codeword $\hat{x}^n(w)$ that is jointly typical with x^n .

② 10.7 CHARACTERIZATION OF THE RATE DISTORTION FUNCTION

$$R(D) = \min_{q(\hat{x}|x): \sum_{\hat{x}} p(x) q(\hat{x}|x) d(x, \hat{x}) \leq D} I(X; \hat{X})$$

$$\text{Set, } J(q) = \sum_x \sum_{\hat{x}} p(x) q(\hat{x}|x) \log \frac{q(\hat{x}|x)}{q(\hat{x})} + \lambda \sum_x \sum_{\hat{x}} p(x) q(\hat{x}|x) d(x, \hat{x}) + \sum_x \nu(x) \sum_{\hat{x}} q(\hat{x}|x)$$

$$(\text{where } q(\hat{x}) = \sum_x p(x) q(\hat{x}|x))$$

$$\text{Diff w.r.t } q(\hat{x}|x). \text{ Set } \log \mu(x) = \nu(x)/p(x) \Rightarrow p(x) \left[\log \frac{q(\hat{x}|x)}{q(\hat{x})} + \lambda d(x, \hat{x}) + \log \mu(x) \right] = 0$$

(Use Lagrange multipliers)

$$\Rightarrow q(\hat{x}|x) = \frac{q(\hat{x}) e^{-\lambda d(x, \hat{x})}}{\mu(x)}$$

$$\text{Since } \sum_{\hat{x}} q(\hat{x}|x) = 1, \mu(x) = \sum_{\hat{x}} q(\hat{x}) e^{-\lambda d(x, \hat{x})} \Rightarrow q(\hat{x}|x) = \frac{q(\hat{x}) e^{-\lambda d(x, \hat{x})}}{\sum_{\hat{x}} q(\hat{x}) e^{-\lambda d(x, \hat{x})}}$$

$$\Rightarrow \times p(x), \text{ summing over all } x \quad q(\hat{x}) = q(\hat{x}) \sum_x \frac{q(\hat{x}) e^{-\lambda d(x, \hat{x})}}{\sum_{\hat{x}} q(\hat{x}) e^{-\lambda d(x, \hat{x})}}$$

(30) If $q(\hat{x}) > 0 \Rightarrow \sum_x \frac{q(\hat{x}) e^{-\lambda d(x, \hat{x})}}{\sum_{\hat{x}'} q(\hat{x}') e^{-\lambda d(x, \hat{x}')}} = 1$

$\Rightarrow$ calculate λ and the $|\hat{X}|$ unknowns $q(\hat{x})$

$\Rightarrow$ find the optimum conditional distribution

The above analysis is valid if $q(\hat{x}) > 0$

$q(\hat{x}) > 0$ is covered by the Kuhn-Tucker conditions, which reduce to

$$\begin{aligned} \frac{\partial J}{\partial q(\hat{x}|x)} &= 0 & \text{if } q(\hat{x}|x) > 0 \\ &\geq 0 & \text{if } q(\hat{x}|x) = 0 \end{aligned}$$

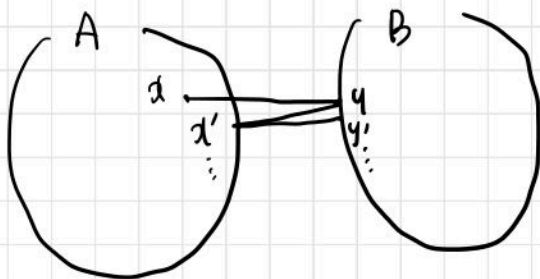
$$\begin{aligned} \Rightarrow \sum_x \frac{q(\hat{x}) e^{-\lambda d(x, \hat{x})}}{\sum_{\hat{x}'} q(\hat{x}') e^{-\lambda d(x, \hat{x}')}} &= 1 & \text{if } q(\hat{x}) > 0 \\ &\leq 1 & \text{if } q(\hat{x}) = 0. \end{aligned}$$

However, it is not easy to solve for the optimum output distribution from these eqs.

31) 10.8 COMPUTATION OF CHANNEL CAPACITY AND THE RATE DISTORTION FUNCTION

o Consider two convex sets A and B in $\mathbb{R}^n$

We would like to find the minimum distance between them



Algorithm

- ① Take any point $x \in A$
- ② find $y \in B$ that is closest to it.
- ③ find $x' \in A$ "
- $\vdots$

[Csiszar and Tusnady] have shown that if convex, certain dist.

$A, B \Rightarrow$ probability distribution
dist $\Rightarrow$ relative entropy

$\Rightarrow$ algorithm does converge to the minimum relative entropy between the two sets of distributions

② To apply this algorithm to rate distortion,

rate distortion function as a minimum of the relative entropy between two sets.

Lemma 10.8.1) $p(x)p(y|x)$: given joint distribution.

distribution $r(y)$ that minimizes the relative entropy $D(p(x)p(y|x) \| p(x)r(y))$ is the marginal distribution $r^*(y)$ corresponding to $p(y|x)$:

$$D(p(x)p(y|x) \| p(x)r^*(y)) = \min_{r(y)} D(p(x)p(y|x) \| p(x)r(y))$$

where $r^*(y) = \sum_x p(x)p(y|x)$.

Also,

$$\max_{r(x|y)} \sum_{x,y} p(x)p(y|x) \log \frac{r(x|y)}{p(x)} = \sum_{x,y} p(x)p(y|x) \log \frac{r^*(x|y)}{p(x)}$$

where $r^*(x|y) = \frac{p(x)p(y|x)}{\sum_x p(x)p(y|x)}$

(33) proof)

$$\begin{aligned} & D(p(x)p(y|x) \| p(x)r(y)) - D(p(x)p(y|x) \| p(x)r^*(y)) \\ &= \sum_{x,y} p(x)p(y|x) \log \frac{p(x)p(y|x)}{p(x)r(y)} - \sum_{x,y} p(x)p(y|x) \log \frac{p(x)p(y|x)}{p(x)r^*(y)} \\ &= \sum_{x,y} p(x)p(y|x) \log \frac{r^*(y)}{r(y)} = \sum_y r^*(y) \log \frac{r^*(y)}{r(y)} = D(r^* \| r) \geq 0 \end{aligned}$$

Using this lemma to rewrite the minimization in the def of the rate distortion function as a double minimization.

$$R(D) = \min_{r(\hat{x})} \min_{q(\hat{x}|x) : \int p(x)q(\hat{x}|x)dx, \hat{x} \leq D} \sum_x \sum_{\hat{x}} p(x)q(\hat{x}|x) \log \frac{q(\hat{x}|x)}{r(\hat{x})}$$

A: Set of all jointly distribution with marginal $p(x)$ that satisfy the distortion constraints

B: Set of product distribution $p(x)r(\hat{x})$ with arbitrary $r(\hat{x})$

$$\Rightarrow R(D) = \min_{r \in B} \min_{p \in A} D(p \| r)$$

MA Blahut - Arimoto algorithm (process of alternating minimization)

① choice λ , initial output distribution $r(\hat{x})$

② calculate the $q(\hat{x}|\lambda)$ that minimizes the mutual information subject to the distortion constraint.

Use Lagrange multiplier

$$q(\hat{x}|\lambda) = \frac{r(\hat{x}) e^{-\lambda d(x, \hat{x})}}{\sum_{\hat{x}} r(\hat{x}) e^{-\lambda d(x, \hat{x})}}$$

③ For this $q(\hat{x}|\lambda)$, calculate $r(\hat{x})$ that minimizes the mutual information, which by Lemma

$$r(\hat{x}) = \sum_{\lambda} p(\lambda) q(\hat{x}|\lambda)$$

④ use this $r(\hat{x})$ as the starting point of the next iteration.

$\Rightarrow$ There is a limit, and the limit has been shown to be $R(D)$ (by Csiszár)
where the value of D and $R(D)$ depends on λ .

$\Rightarrow$ Choosing λ appropriately sweeps out the $R(D)$ curve.

(25) A similar procedure can be applied to the calculation of channel capacity

$$C = \max_{r(x)} I(X;Y) = \max_{r(x)} \sum_x \sum_y r(x) p(y|x) \log \frac{r(x) p(y|x)}{r(x) \sum_{x'} r(x') p(y|x')}$$

Using
Lemma 10.8.1

$$\Rightarrow q(x|y) = \frac{r(x) p(y|x)}{\sum_x r(x) p(y|x)} \quad (\text{by Lemma 10.8.1})$$

$$\Rightarrow r(x) = \frac{\pi_y(q(x|y)) p(y|x)}{\sum_x \pi_y(q(x|y)) p(y|x)}$$

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These algorithm for the computation of the channel capacity and the rate distortion function were established by Blahut and Arimoto and the convergence for the rate distortion computation was proved by Csiszar.

The alternating minimization procedure of Csiszar and Tusnady can be specialized to many other situations as well, including the EM algorithm, and the algorithm for finding the log-optimal portfolio for a stock market.

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Thank you!